



UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Course Syllabus – Fall 2024
Course Name: Fundamental Calculus Math -1151

1	Course Title	Basic Calculus (Differential and Integral Calculus)													
2	Course Code	Math-1151													
3	Trimester and Year	Spring 25													
4	Pre-requisites	N/A													
5	Credit Hours	3													
6	Section	H													
7	Class Hours	Sun and Wed: 11:11 AM - 12:30 PM													
8	Class Room	Room No# 804													
9	Instructor's Name	Dr. Jashodhan Saha													
10	Email	jsaha@ins.uiu.ac.bd , Mobile: 01712204400													
11	Office	Room No# 735 (B)													
13	Textbook	Calculus Early Transcendentals, Howard Anton, Irl Bivens, Stephen Davis [10th Edition]													
14	Reference	Engineering Mathematics, H. K. Dass (HKD) [15th Edition]													
15	Course Contents (approved by UGC)	Function, Domain and Range of a Function, Translation, reflection, compression and stretches of a function. Even and Odd functions, Inverse functions, One to One and many to one function, Family of Exponential, logarithmic, sine and cosine function, Limit, continuity and differentiability, Tangent line, Derivative and Chain rule, An overview of area problem, Newton's anti-derivative method in finding area, Indefinite integral, fundamental theorem of calculus, Definite integral, Area between two curves, arc length.													
16	Course Outcomes (COs)	<table><tr><th>Cos</th><th>Description</th></tr><tr><td>CO1</td><td>Define relation of two variables, interval, function, domain and range of different types of function, Composite, odd and even functions, inverse functions, family of functions, Limit, continuity and differentiability of various functions.</td></tr><tr><td>CO2</td><td>Analyze graph of functions (Translations, reflections, Stretch and compress), limits, continuity and differentiability.</td></tr><tr><td>CO3</td><td>Compute differentiations and integrations of functions.</td></tr><tr><td>CO4</td><td>Apply differentiation to find the rate of change of functions, velocity of a moving particle, average velocity, instantaneous velocity and slope of secant and tangent line.</td></tr><tr><td>CO5</td><td>Interpret the relationship of definite integral with area problem, indefinite integral and arc length.</td></tr></table>		Cos	Description	CO1	Define relation of two variables, interval, function, domain and range of different types of function, Composite, odd and even functions, inverse functions, family of functions, Limit, continuity and differentiability of various functions.	CO2	Analyze graph of functions (Translations, reflections, Stretch and compress), limits, continuity and differentiability.	CO3	Compute differentiations and integrations of functions.	CO4	Apply differentiation to find the rate of change of functions, velocity of a moving particle, average velocity, instantaneous velocity and slope of secant and tangent line.	CO5	Interpret the relationship of definite integral with area problem, indefinite integral and arc length.
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17	Teaching Methods	Lecture, Case Studies, Assignments.													

	Equipment & Aids	Bring your own device (Any standard smartphone or tablet or laptop) to participate effectively in classroom activities. You are not allowed to borrow from others inside the classroom during class activities																																																																																																				
	Course Rationale	The use of Calculus is widespread in science, engineering, business, and many other fields. Calculus is a tool used almost everywhere in the modern world to describe change and motion. It also provides important tools in understanding functions and has led to the development of new areas of mathematics including real and complex analysis and geometry. This course is a prerequisite to higher courses																																																																																																				
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				Chap: 2.2 Exercise-47,48. Chap: 2.1 Exercise:11 – 18	
			MIDTERM EXAM		
	13-14	Differentiation of different types of functions (Derivative and Chain rule)	3	Chap: 2.6 Exercise: 7 – 49 (odd)	Q/A Test, Assignment
	15	An overview of area problems, Newton's anti-derivative method in finding area,	4	Chap: 5.1 Example: 1, 2; exercise: 13 – 18	Q/A Test, Assignment
	16-20	Indefinite integral, fundamental theorem of calculus, Definite integral, Area between two curves,	4,5	Chap: 5.5 Exercise: 13 – 20 Chap: 5.2 Exercise: 5 – 8, 15 – 30. Chap: 5.3 Exercise: 15 – 56 Chap: 5.6 Exercise:13 – 30, Chap: 6.1 Exercise:1 – 7, 11 – 12	Q/A Test, Assignment
	21-23	Different types of Integration (Principles of Integral evaluation, Integration by parts, Trigonometric Substitution).	6	7.1: A review of familiar integration formulas: 1-13, 21-26 Exercise Set 7.1: 1-30 7.2: Examples: 1-8 Exercise Set 7.2: 1-38 7.4: Examples: 1-6 Exercise Set 7.4: 1-26, 37-47 (Only odd numbers)	Q/A Test
	24	Review Class			

Appendix 1: Assessment Methods

Assessment Types	Marks
Attendance	5%
Assignments	5%
Class Tests	20%
Mid Term	30%
Final Exam	40%

Appendix 2: Grading Policy

Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix-3: Program outcomes

POs	Program Outcomes
PO1	An ability to apply knowledge of mathematics in science, and engineering
PO2	An ability to identify, formulate, and solve engineering problems
PO3	An ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability
PO4	An ability to design and conduct experiments, as well as to analyze and interpret data
PO5	An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice
PO6	The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
PO7	A knowledge of contemporary issues
PO8	An understanding of professional and ethical responsibility
PO9	An ability to function on multidisciplinary teams
PO10	An ability to communicate effectively
PO11	Project Management and Finance
PO12	A recognition of the need for, and an ability to engage in life-long learning